

26 Confidence Intervals

- The estimators we have considered so far are *point* estimators; we estimate the population parameter θ , an unknown number, by the observed sample statistic $\hat{\theta}$, a number which can be computed based on sample data $X_1, \dots, X_n$
- For a particular sample a particular numerical value of $\hat{\theta}$ is observed, but there are many possible randomly selected samples and many possible values of the random variable $\hat{\theta}$.
- While for any particular sample we hope the observed value of $\hat{\theta}$ is close to the true value of θ , there will be some error in estimation due to natural sample-to-sample variability.
- Because of this natural sample-to-sample variability, it is usually better to provide a range of values when estimating a parameter
- In determining a plausible range of values, there is a trade-off between
 - Accuracy: the wider the interval the more confident we are that it actually contains the true parameter value
 - Precision: but the wider the interval the less precise of an estimate it provides
- The usual approach is to specify a desired degree of accuracy. Specifically, interval estimate procedures are constructed so that they will be accurate “most of the time”.
- A *confidence interval* estimates a population parameter based on sample data with
 - a plausible range of values and
 - a *confidence level* that the estimated range contains the true parameter value.
 - * By far, the most commonly used confidence level is 95%.
 - * But see Exercise 26.2 and related material about multiple confidence intervals

Example 26.1. In a [recent survey](#) of a representative sample of U.S. adults aged 18-29, 67% saw a movie in the theater in the last 12 months. The survey reports a margin of error for 95% confidence of 4.2 percentage points.

1. Use the information provided to compute and interpret a 95% confidence interval.

2. The study also surveyed 3257 U.S. adults aged 30-49, and 60% saw a movie in the theater in the last year. Without computing, will the margin of error for 95% confidence for the 30-49 age group be greater than, less than, or equal to 4.2 percentage points?

3. For the 30-49 age group the margin of error for 95% confidence is 2.2 percentage points. Compute and interpret a 95% confidence interval.

- A $1 - \alpha$ **confidence interval** for θ is an *interval estimator* of θ based on the sample $X_1, \dots, X_n$ of the form $[L(X_1, \dots, X_n), U(X_1, \dots, X_n)]$ which satisfies

$$P(L(X_1, \dots, X_n) \leq \theta \leq U(X_1, \dots, X_n)) = 1 - \alpha$$

- $L(X_1, \dots, X_n)$ and $U(X_1, \dots, X_n)$ represent, respectively, the lower and upper endpoints of the interval. L and U are *random variables* that vary from sample-to-sample.
- Typical values of α include 0.05 and 0.01. We express $1 - \alpha$ as a percentage. For example, $\alpha = 0.05$ corresponds to a 95% confidence interval.
- Often $L = \hat{\theta} - M$ and $U = \hat{\theta} + M$, where the **margin of error** M — where M can be a RV — accounts for both the sample-to-sample variability of $\hat{\theta}$ and the confidence level.
 - The margin of error is often of the form: multiple of standard error
 - The multiple is determined by the confidence level and the distribution of the point estimator $\hat{\theta}$ (e.g., 2 for 95% confidence when the estimator follows a Normal distribution.)
 - * The larger the confidence level, the larger the margin of error — and the wider the range of plausible values — required to achieve that level of confidence.
 - The **standard error** is the estimated sample-to-sample SD of the estimator, $SE(\hat{\theta}) = \widehat{SD}(\hat{\theta})$.
 - * The less variable the statistic (from sample to sample) the smaller the margin of error and the narrower the interval.
 - * In general, the larger the sample size, the smaller the SE of the statistic, the smaller the margin of error, and the narrower the interval.

Example 26.2. Consider the general problem of estimating a population mean μ based on a random sample of size n . A natural point estimator of μ is the sample mean $\bar{X}$. The

central limit theorem says that, if n is large enough, $\bar{X}$ has an approximate $\text{Normal}(\mu, \sigma/\sqrt{n})$ distribution, where σ is the population standard deviation. We'll use this theory to derive a confidence interval for μ .

1. Fill in the blank: for 95% of samples, the sample mean $\bar{X}$ is within [blank] of the population mean μ .
2. Fill in the blank: for 95% of samples, the interval with endpoints $\bar{X} \pm$ [blank] will contain the population mean μ .
3. The previous part suggests a formula for a CI for μ . However, this is not a practically useful formula; explain why. Then suggest how we can fix it.
4. Everything else being equal, how does the sample size affect the confidence interval and its margin of error: the larger the sample size...? Does this make sense?
5. Everything else being equal, how does the confidence level (e.g. 95%) affect the confidence interval and its margin of error: the larger the confidence level...? Does this make sense?

- A one-sample t confidence interval for a population mean μ has endpoints

$$\bar{X} \pm t^* \frac{S}{\sqrt{n}}$$

where S is the sample standard deviation $S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$ and t^* , the multiple corresponding to the confidence level, is from the t distribution with $n - 1$ degrees of freedom.

- For all practical purposes you can treat t distributions like Normal distributions and use the usual empirical rule
 - Technically, t^* is a little bigger than z^* to account for the additional sample-to-sample variability in S .
 - But in most situations encountered in practice, t^* and z^* are essentially the same.
- Assumptions:
 - Simple random sample from a large population, that is, independent observations
 - No bias in data collection
 - Population mean is the parameter you want to estimate

Confidence level	90%	95%	99%	99.7%	99.99%
Multiple ($z^* \approx t^*$)	1.7	2.0	2.6	3.0	4.0

Example 26.3. A frequently used example data set is the [Ames housing data set](#) which consists of residential properties sold in Ames, Iowa from 2006 to 2010. For more information about the variables in this data set, refer to the [data documentation](#).

Suppose we wish to use this data set to estimate μ , the *population mean sale price of single family homes*.

In the sample, there are 2425 single family homes with a sample mean sale price (\$K) of 185 and sample standard deviation 83.

1. Compute a 95% confidence interval.
2. Write a clearly worded sentence containing the conclusion of the confidence interval in context.

3. To what extent can you generalize your results? That is, what is the relevant “population”? What are some issues to consider?

 4. What if we wanted 95% confidence, but we wanted the margin of error to be no bigger than 1K. How could we achieve this?

 5. Assuming that this is a random sample from the appropriate population, does the 95% confidence interval you computed in the previous part provide an accurate estimate of the corresponding parameter? That is, is μ contained within the CI? Explain.
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- Because the value of the parameter is *unknown*, we never know for sure whether any one confidence interval contains the parameter or not.
 - But the procedure guarantees that in the long run — that is, over many unbiased random samples — “most” confidence intervals will contain the parameter, regardless of what the true value of the parameter is. “Most” is determined by the level of confidence.
 - Any value within the confidence interval is a plausible value of the parameter
 - Values outside of the confidence interval are not plausible values of the parameter
 - The CI could be wrong; we are confident that it is correct, but we’ll never know for sure
 - There are two sources of differences between a statistic and a corresponding parameter
 - systematic errors due to **bias**
 - “chance errors” due to **sampling variability**
 - While there is no bias inherent in simple random sampling from a population, bias can be present in other ways, including undercoverage, nonresponse, wording of questions, etc.
 - Bias concerns the *center* of the sampling distribution of a statistic. A biased sampling procedure produces values which consistently overestimate or consistently underestimate the respective parameter.

- The inference procedures — confidence intervals and hypothesis tests — we will encounter assume that the data are collected in an unbiased way from a random sample.
- If there is bias present in data collection then the observed statistic (i.e. the center of the confidence interval) is likely to be a poor estimate of the parameter.
- The margin of error in a confidence interval only accounts for sampling variability, NOT bias.
- To see if bias is present — in which case the confidence interval cannot be trusted — ask: how was the data collected?

Example 26.4. Continuing Example [26.3](#).

1. Consider the distribution of sale price. What shape do you expect this distribution to have? Why?
 2. A student says, “The distribution of sale price is clearly skewed to the right. Since it’s not Normal the confidence interval we computed, and the resulting conclusion about the population mean, is not valid.” Do you agree with this critique? Explain.
 3. Considering that the distribution of sale price is skewed to the right, suggest a valid criticism of the confidence interval we computed. Hint: what parameter is it estimating?
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- The t interval for the population *mean* is valid as long as the sample size is large enough, regardless of the shape of the population distribution.
 - However, if the population distribution is highly skewed, then the population *median* might be a more appropriate parameter to consider as a measure of center.
 - Which is more appropriate, the population mean or the population median, depends on the purpose for the inference.
 - ALWAYS PLOT YOUR DATA, and consider the distribution of the variable. If the values in a representative sample are skewed or include large outliers, the same is also plausibly true for the population it’s supposed to represent.

26.1 Exercises

Exercise 26.1. For each of the following statements, identify if it is a valid interpretation of the confidence interval from Example 26.3. If not, explain why.

1. We estimate that 95% of single family homes in the population have sale price (\$K) between 181 and 188.
 2. We estimate with 95% confidence that the sample mean sale price (\$K) of single family homes is between 181 and 188.
 3. If many samples of this size were selected about 95% would produce a CI of (181, 188).
 4. If many hypothetical unbiased random samples of this size were selected, and a 95% confidence interval computed based on each sample, about 95% of these confidence intervals would contain the population mean sale price.
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- A CI is a conclusion about a population parameter — a number that *summarizes* the population (e.g. the population mean) — and not about individual values of a variable.
 - The conclusions from inference regard the *population* from which the random sample was selected; there is nothing to conclude for the sample.
 - No one particular sample (e.g. the one observed) or CI is necessarily good or representative of the many other possible samples or CIs.
 - The randomness is in the *sampling procedure*, which determines L and U , and not the parameter. The parameter θ is a fixed number. After selecting the sample, there is no

randomness left. Either the CI contains the parameter or not; you just don't know which it is.

- We are confident in our observed estimate because we are confident in our estimation *procedure*
 - By considering what would happen over many hypothetical random samples we are able to determine the necessary margin of error so that our CIs would contain the parameter in 95% of samples.
 - Because the value of the parameter is unknown, we never know for sure whether any one confidence interval contains the parameter or not.
 - But the procedure guarantees that in the long run — that is, over many unbiased random samples — “most” confidence intervals will contain the parameter, regardless of what the true value of the parameter is. “Most” is determined by the level of confidence.

Exercise 26.2. Suppose we wish to estimate population mean GPA of students at each of the 23 CSU campuses. For each campus, you select a random sample of students, collect the GPAs of the students, and then use the sample data to compute a 95% confidence interval. That is you compute 23 separate 95% confidence intervals, based on 23 independent random samples, for each of 23 population means.

1. First just consider Cal Poly. What is the probability that your confidence interval based on the Cal Poly sample contains the true population mean GPA at Cal Poly?

2. What is the probability that all 23 of your 95% confidence intervals contain their respective population mean?

3. How confident are you that all 23 of your 95% confidence intervals contain their respective population mean?

- When computing multiple confidence intervals, the multiple (z^* or t^*) of each confidence interval should be increased to account for the **issue of multiple comparisons**.
 - For example, with a multiple of 2 in a *single* confidence interval, we are 95% confident that the interval contains the corresponding parameter.
 - However, if we use a multiple of 2 for each of 10 confidence intervals, we are only 60% confident¹ that *all* of the 10 95% confidence intervals contain their respective parameters.
 - A **Bonferroni** adjustment, splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the z^*/t^* multiple for 99.5% confidence when constructing each interval.
 - A Bonferroni adjustment guarantees **simultaneous confidence**
 - But it is conservative, and tends to produce wider intervals than necessary
 - There are other methods; for example, the *Tukey-Kramer method* is often used for *all pairwise comparisons of means* (e.g., in ANOVA)
4. Suppose we want simultaneous 95% confidence that all 23 CIs contain their respective mean. Find the z^* multiple corresponding to a Bonferroni adjustment. How many times wider will each CI be than the unadjusted intervals?
5. How confident are you that all 23 of your adjusted confidence intervals contain their respective population mean?

Exercise 26.3. Continuing with the Ames housing data set. Now suppose we want to compare sale price of single family homes with other homes (including condos, townhouses, etc). In particular, suppose we wish to estimate $\mu_S - \mu_N$, the difference in population mean sale price between single family homes and non-single family homes. The table below summarizes the sample data.

¹This number assumes independence of confidence intervals — $0.95^{10} \approx 0.60$ — but similar ideas apply more generally.

	count	mean	std	min	25%	50%	75%	max
SingleFamily								
False	505.0	161.511398	60.394023	55.000	120.0	147.4	190.0	392.5
True	2425.0	184.812041	82.821802	12.789	130.0	165.0	220.0	755.0

1. Suggest a point estimator, compute its variance, and suggest a formula for a CI for $\mu_S - \mu_N$.

2. Compute a 95% confidence interval.

3. Write a clearly worded sentence interpreting the confidence interval from the previous part in context.

4. Are we reasonably confident that single family homes tend to have higher sale prices than other homes? Explain.

- For many statistics (e.g. means, differences in means, proportions, differences in proportions) the sample-to-sample pattern of variability of values of the statistic over many unbiased random samples has an approximate Normal shape centered at the true value of the population parameter

- In such situations confidence intervals take the form

$$\text{statistic} \pm \text{multiple of its SE}$$

- If two samples are selected *independently* from each of two populations with parameters θ_1 and θ_2 , then an approximate $1 - \alpha$ CI for $\theta_1 - \theta_2$ has endpoints

$$\hat{\theta}_1 - \hat{\theta}_2 \pm t^* \text{SE}(\hat{\theta}_1 - \hat{\theta}_2)$$

where

$$\text{SE}(\hat{\theta}_1 - \hat{\theta}_2) = \sqrt{\text{SE}(\hat{\theta}_1)^2 + \text{SE}(\hat{\theta}_2)^2}$$

- The above assumes that $\hat{\theta}_1$ and $\hat{\theta}_2$ each have an approximate Normal distribution, and so, since they are independent, $\hat{\theta}_1 - \hat{\theta}_2$ also has an approximate Normal distribution.
- A difference provides an absolute comparison between groups.
 - Ratios can be used to provide relative comparisons; however, the CI formulas are more complicated, but we'll see another approach soon

Exercise 26.4. You'll use a [simulation applet](#) that randomly generates confidence intervals for a population mean to help you understand some ideas. The applet calculates a confidence interval for each set of randomly generated data.

- In the box for “Statistic” select means “Means”.
- In the box next to “Method” select “t”.
- Start with “Number of intervals” equal to 1 at first to see what is happening, but then change it to lots.

Experiment with different simulations; be sure to change

- Distribution
- Population Mean
- Population SD
- Sample size
- Confidence level

Then write a paragraph or two or some bullet points of your main observations. Based on this applet, what can you say about how confidence intervals work? How does changing each of the inputs affect the confidence intervals and the Results? Include a few screenshots from the applet to support your conclusions.